

Week 3: Coding Day

“Teamworking and Collaborative Learning”

Instruction

- Student must complete the **Week 3 Coding Day Challenges** on **USART and I2C** for exam revision and the AMR project development.
- Develop STM32F446 code to acquire real-time data from the **RPLidar C1** and **MPU6500** using **interrupts, DMA, and relevant peripheral register configuration**.
- Use **PulseView** to analyse the sensor **signals and protocols**, and **convert** captured data into **meaningful physical values**.
- The **minimum required analysis** is the revision for **final closed-book examination** reviewion, namely RPLidar C1 SCAN mode and MPU6500 gyroscope transactions.
- Additional extension work should include **RPLidar C1 Express mode** and the **MPU6500** accelerometer are required for the AMR project development.
- Submit an **analysis report** and demonstrate the working system to the **TA**. The **Coding Day mark** will be awarded **individually** through questioning based on the report, the protocol analysis, and the live demonstration



Submit report
using this QR
code

Task 1: STM32F446 Real-Time Sensor Interfacing

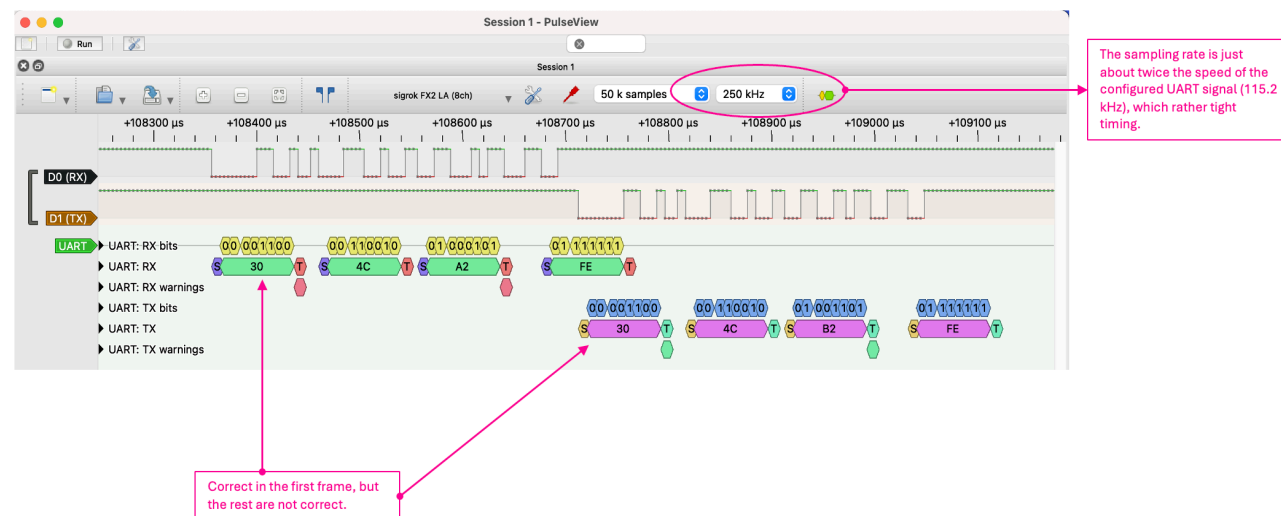
- Develop an integrated STM32F446 code to acquire data from:
 - RPLidar C1 using USART
 - MPU6500 using I2C
- Your implementation must use a **real-time interrupt and DMA approach**, not simple polling. This includes configuring the relevant peripheral registers as appropriate, such as:
 - USART configuration registers for baud rate and communication setup
 - I2C configuration registers for master communication
 - DMA stream/channel configuration
 - interrupt enable and NVIC configuration
- Your code should demonstrate that the microcontroller can continuously receive or transfer sensor data reliably in real time.

Task 2: Protocol Analysis Using PulseView

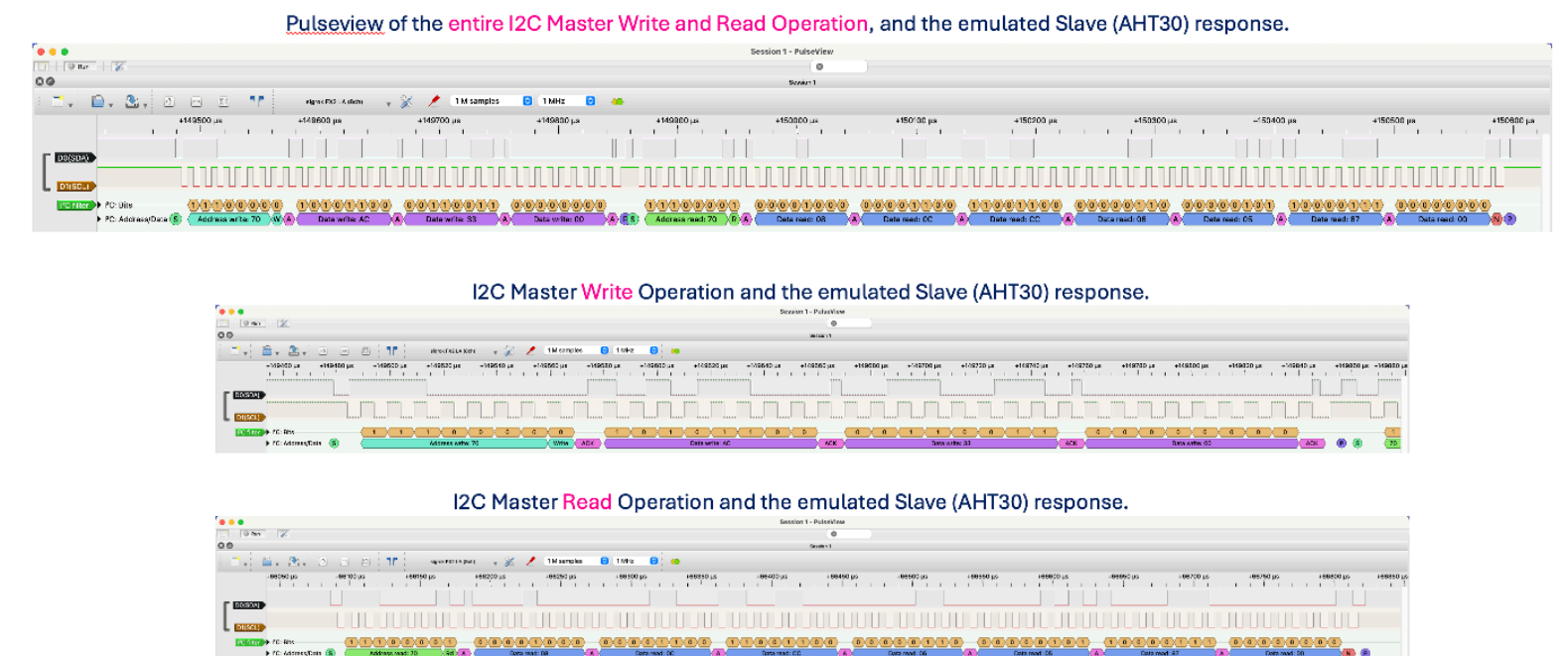
- Using **PulseView logic analyser** to capture and analyse the signalling and protocol communication of both RPLidar C1 and MPU6500 sensors.
- Your analysis must include the **minimum required parts** listed in the following sub-sections of RPLidar C1 and MPU6500, which are based directly on the knowledge required for **final closed-book examination**, together with additional extension analyses towards the project development.

Pulseview Signaling and Protocol Analyses

■ USART with DMA: UART signal analysis with Pulseview



■ I2C protocol analysis: Master and Slave (AHT30) Communication



Task 2.1: RPLidar C1 Analysis

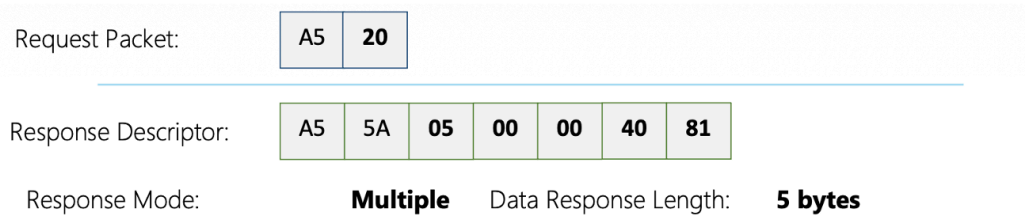
- Capture and analyse the UART communication between the STM32F446 and the RPLidar C1.
- Minimum required analysis
 - Configure the USART communication settings used for the sensor
 - Identify the response descriptor in SCAN mode
 - Identify the start of each measurement sample
 - Explain how the start bits work in the SCAN-mode packet format
 - Decode at least one measurement sample
 - Determine and explain:
 - whether the sample is valid
 - the quality value
 - the angle in degrees
 - the distance in mm
- Additional extension analysis
 - Express mode packet structure and behaviour;
 - How Express mode differs from normal SCAN mode
 - How the captured bytes would be interpreted in Express mode

Task 2.1: RPLidar C1 Analysis

Start Scan (SCAN) Request and Response

Note: The sampling rate of S Series products will be lower when processing this request. Please use EXPRESS_SCAN for the best performance.

- Traditional Mode Protocol (Standard)



RPLIDAR, except for the RPLIDAR that is in the Protection Stop State, will enter the scanning state once it receives this request from a host system. Each measurement sample result will be sent out using an individual data response packet. If the RPLIDAR has been in scanning state already, it will stop the current measurement sampling and start a new round of scanning. This request will be ignored when RPLIDAR is in the Protection Stop state.

The related response descriptor will be sent out by RPLIDAR immediately once it receives the request and accepts it. The data response packets related to every measurement sample results will be sent out continuously only after the motor rotation becomes stable. RPLIDAR will leave the Scanning state once it receives a new request from the host system or it detects something is wrong.

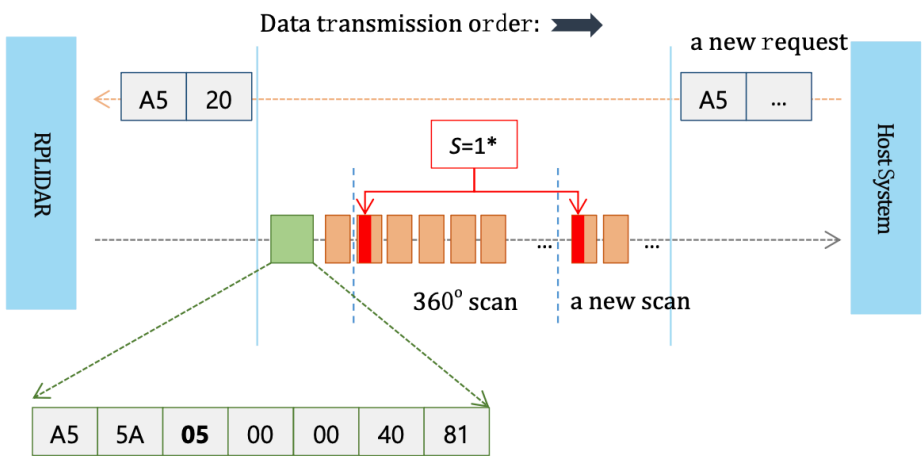


Figure 4-10 The Communication Status after Host System Sending a SCAN Request

Format of the Data Response Packets :

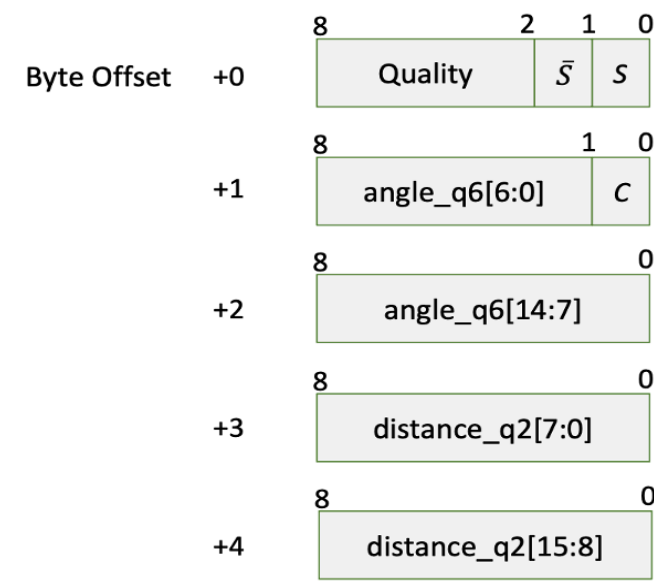


Figure 4-4 Format of a RPLIDAR Measurement Result Data Response Packet

RPLIDAR encapsulates each measurement sample into a data response packet with the format showed in the above figure and send the packet out. The descriptions of every field within the packet are listed in the following table:

Field Name	Description	Examples / Notes
S	Start flag bit of a new scan	When S is set to 1, the current and incoming packets belong to a new 360° scan.
$\bar{S}$	Inversed start flag bit, always has $\bar{S} = !S$	Can be used as a data check bit.
C	Check bit, constantly set to 1	Can be used as a data check bit.
quality	Quality of the current measurement sample	Related the reflected laser pulse strength.
angle_q6	The measurement heading angle related to RPLIDAR's heading. In degree unit, [0-360) Stored using fix point number.	Refer to the below figure for details. Actual heading = angle_q6/64.0 Degree
distance_q2	Measured object distance related to RPLIDAR's rotation center. In millimeter (mm) unit. Represents using fix point. Set to 0 when the measurement is invalid.	Refer to the below figure for details. Actual Distance = distance_q2/4.0 mm

Figure 4-5 Field Definition of a RPLIDAR Measurement Result Data Response Packet

$$\text{AngleDiff}(\omega_i, \omega_{i+1}) = \begin{cases} \omega_{i+1} - \omega_i, & \omega_i \leq \omega_{i+1} \\ 360 + \omega_{i+1} - \omega_i, & \omega_i > \omega_{i+1} \end{cases}.$$

Task 2.2: MPU6500 Analysis

- Capture and analyse the I2C communication between the STM32F446 and the MPU6500.
- Minimum required analysis
 - Identify the device address and the meaning of the read/write address bytes
 - Identify the register address used in the transaction
 - Explain the I2C transaction sequence, including:
 - START, address + write, register selection, repeated START, address + read, returned data bytes, ACK/NACK and STOP
 - Reconstruct the returned gyroscope values from the captured bytes
 - Convert the raw gyroscope readings into:
 - 16-bit hexadecimal values
 - signed decimal values
 - physical values in appropriate units
- Additional extension analysis
 - the accelerometer registers and read sequence
 - reconstruction of raw accelerometer output data
 - conversion of accelerometer readings into meaningful physical values

Task 2.2: MPU6500 Analysis



MPU-6500 Register Map and Descriptions

Document Number: RM-MPU-6500A-00
Revision: 2.1
Release Date: 09/16/2013

41	65	TEMP_OUT_H	R	TEMP_OUT_H[15:8]
42	66	TEMP_OUT_L	R	TEMP_OUT_L[7:0]
43	67	GYRO_XOUT_H	R	GYRO_XOUT_H[15:8]
44	68	GYRO_XOUT_L	R	GYRO_XOUT_L[7:0]
45	69	GYRO_YOUT_H	R	GYRO_YOUT_H[15:8]
46	70	GYRO_YOUT_L	R	GYRO_YOUT_L[7:0]
47	71	GYRO_ZOUT_H	R	GYRO_ZOUT_H[15:8]
48	72	GYRO_ZOUT_L	R	GYRO_ZOUT_L[7:0]

4.24 Registers 67 to 72 – Gyroscope Measurements

Name: GYRO_XOUT_H

Serial IF: READ

Reset value: 0x00 (if sensor disabled)

BITS	NAME	FUNCTION
[7:0]	GYRO_XOUT_H	High byte of the X-axis gyroscope output

Name: GYRO_XOUT_L

Serial IF: READ

Reset value: 0x00 (if sensor disabled)

BITS	NAME	FUNCTION
[7:0]	GYRO_XOUT_L	Low byte of the X-axis gyroscope output GYRO_XOUT = Gyro_Sensitivity * X_angular_rate Nominal FS_SEL = 0 Conditions Gyro_Sensitivity = 131 LSB/(°/s)

Name: GYRO_YOUT_H

Serial IF: READ

Reset value: 0x00 (if sensor disabled)

BITS	NAME	FUNCTION
[7:0]	GYRO_YOUT_H	High byte of the Y-axis gyroscope output

Name: GYRO_YOUT_L

Serial IF: READ

Reset value: 0x00 (if sensor disabled)

BITS	NAME	FUNCTION
[7:0]	GYRO_YOUT_L	Low byte of the Y-axis gyroscope output GYRO_YOUT = Gyro_Sensitivity * Y_angular_rate Nominal FS_SEL = 0 Conditions Gyro_Sensitivity = 131 LSB/(°/s)

Name: GYRO_ZOUT_H

Serial IF: READ

Reset value: 0x00 (if sensor disabled)

BITS	NAME	FUNCTION
[7:0]	GYRO_ZOUT_H	High byte of the Z-axis gyroscope output

Name: GYRO_ZOUT_L

Serial IF: READ

Reset value: 0x00 (if sensor disabled)

BITS	NAME	FUNCTION
[7:0]	GYRO_ZOUT_L	Low byte of the Z-axis gyroscope output GYRO_ZOUT = Gyro_Sensitivity * Z_angular_rate Nominal FS_SEL = 0 Conditions Gyro_Sensitivity = 131 LSB/(°/s)

Task 2.2: MPU6500 Analysis

	MPU-6500 Register Map and Descriptions	Document Number: RM-MPU-6500A-00 Revision: 2.1 Release Date: 09/16/2013
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3B	59	ACCEL_XOUT_H	R	ACCEL_XOUT_H[15:8]
3C	60	ACCEL_XOUT_L	R	ACCEL_XOUT_L[7:0]
3D	61	ACCEL_YOUT_H	R	ACCEL_YOUT_H[15:8]
3E	62	ACCEL_YOUT_L	R	ACCEL_YOUT_L[7:0]
3F	63	ACCEL_ZOUT_H	R	ACCEL_ZOUT_H[15:8]
40	64	ACCEL_ZOUT_L	R	ACCEL_ZOUT_L[7:0]

4.22 Registers 59 to 64 – Accelerometer Measurements

Name: ACCEL_XOUT_H

Serial IF: READ

Reset value: 0x00 (if sensor disabled)

BIT	NAME	FUNCTION
[7:0]	ACCEL_XOUT_H	High byte of accelerometer X-axis data.

Name: ACCEL_XOUT_L

Serial IF: READ

Reset value: 0x00 (if sensor disabled)

BIT	NAME	FUNCTION
[7:0]	ACCEL_XOUT_L	Low byte of accelerometer X-axis data.

Name: ACCEL_YOUT_H

Serial IF: READ

Reset value: 0x00 (if sensor disabled)

BIT	NAME	FUNCTION
[7:0]	ACCEL_YOUT_H	High byte of accelerometer Y-axis data.

Name: ACCEL_YOUT_L

Serial IF: READ

Reset value: 0x00 (if sensor disabled)

BIT	NAME	FUNCTION
[7:0]	ACCEL_YOUT_L	Low byte of accelerometer Y-axis data.

Name: ACCEL_ZOUT_H

Serial IF: READ

Reset value: 0x00 (if sensor disabled)

BIT	NAME	FUNCTION
[7:0]	ACCEL_ZOUT_H	High byte of accelerometer Z-axis data.

Name: ACCEL_ZOUT_L

Serial IF: READ

Reset value: 0x00 (if sensor disabled)

BIT	NAME	FUNCTION
[7:0]	ACCEL_ZOUT_L	Low byte of accelerometer Z-axis data.

Task 3: Analysis Report

- Prepare a short analysis report that clearly documents:
 - Your peripheral configuration approach
 - How interrupts and DMA are used
 - PulseView captures of UART and I2C transactions
 - Protocol interpretation for each sensor
 - Example calculations showing how raw bytes are converted into meaningful values
 - Brief explanation of any implementation issues, debugging steps, or limitations
- The report should be technical, concise, and structured clearly enough that each member can explain it confidently.

End of Week3